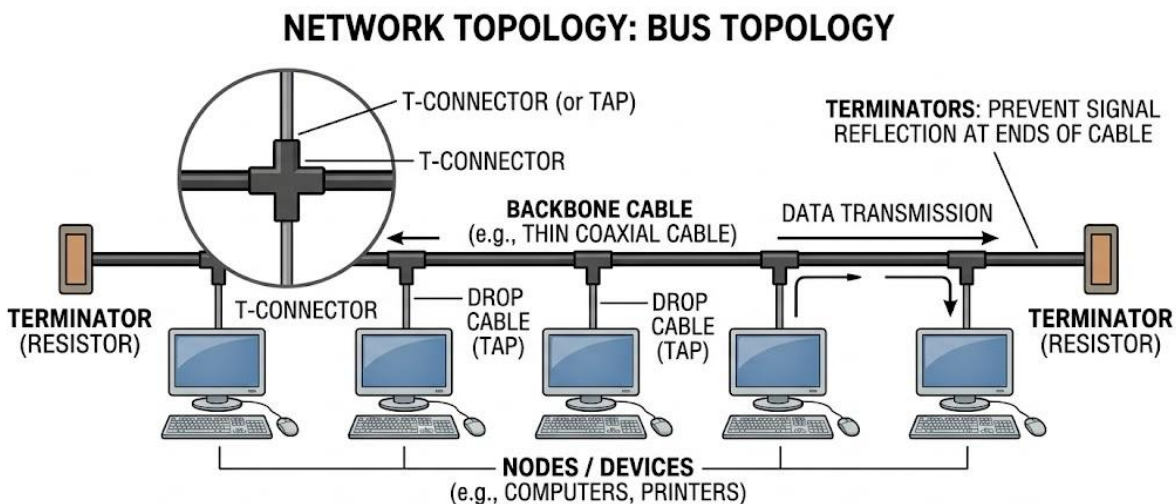


WIRELESS NETWORKING

Day – 7 Network topologies theory.

Task – 1 Bus topologies and labelling.



REAL-WORLD EXAMPLE:

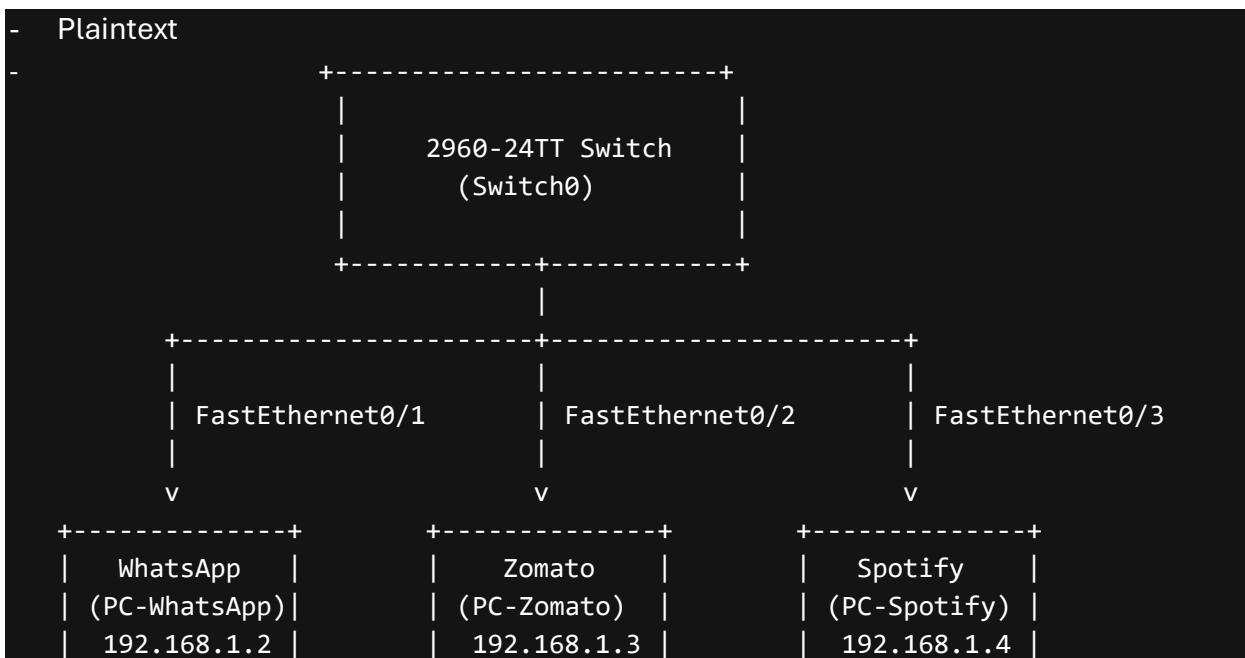
Legacy Ethernet Networks: Pre-1990s 10BASE2 (Thinnet) or 10BASE5 (Thicknet) LANs in early offices or classrooms used a common coaxial backbone to connect a limited number of devices, which was relatively simple and inexpensive to deploy for small networks at the time.

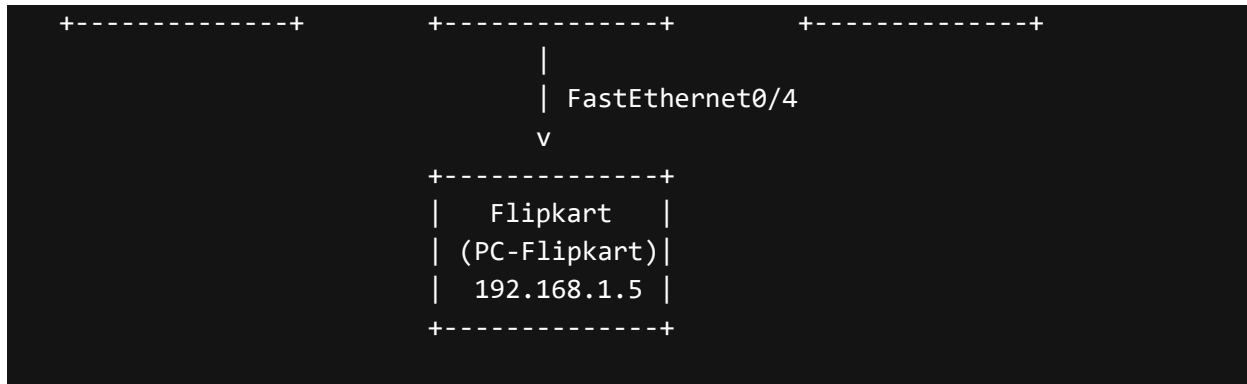
Task – 2 Cisco packet tracer.

- 1.Add the Switch:1 min.
- From the bottom-left device toolbar, select Network Devices > Switches, drag a 2960 Switch onto the workspace, and leave its display name as Switch0.
-
- 2.Add and Rename the 4 PCs:2 min.
- Select End Devices > PC and drag 4 PCs onto the workspace surrounding the switch. Click on each PC's label to rename them:
- PC 1: WhatsApp
- PC 2: Zomato

- PC 3: Spotify
- PC 4: Flipkart
-
- 3.Connect Devices with Straight-Through Cables:2 min.
- Select Connections (the lightning bolt icon) and choose Copper Straight-Through cable (solid black line):
- Connect WhatsApp (FastEthernet0) to Switch0 (FastEthernet0/1).
- Connect Zomato (FastEthernet0) to Switch0 (FastEthernet0/2).
- Connect Spotify (FastEthernet0) to Switch0 (FastEthernet0/3).
- Connect Flipkart (FastEthernet0) to Switch0 (FastEthernet0/4).
-
- 4.Configure IP Addresses:3 min.
- Click on each PC, navigate to Desktop > IP Configuration, set to Static, and enter:
- WhatsApp: IP 192.168.1.2 | Subnet Mask 255.255.255.0
- Zomato: IP 192.168.1.3 | Subnet Mask 255.255.255.0
- Spotify: IP 192.168.1.4 | Subnet Mask 255.255.255.0
- Flipkart: IP 192.168.1.5 | Subnet Mask 255.255.255.0
-
- 5.Capture Screenshot:Instant.
- Wait until all port indicators turn from orange to green. Use Windows Key + Shift + S (or Cmd + Shift + 4 on Mac) to snip your Packet Tracer canvas.

- Cisco Packet Tracer Topology Diagram





Task – 3 Ring topologies.

- Data Transmission in a Ring Topology :- In a ring topology, each device connects directly to two neighboring devices, forming a single continuous circular pathway for data signals. Data travels in one specific direction (unidirectional, usually clockwise) from node to node, where each device inspects the data packet's destination address and retransmits it until it reaches the target machine. Many traditional ring networks use a "token-passing" mechanism, allowing only the device holding the token frame to transmit data and prevent packet collisions.
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Task – 4 Mesh and star topologies comparison.

Topology	Advantages	Disadvantages
Star Topology	• Easy Centralized Management: Adding, removing, or troubleshooting devices is straightforward because all connections go to a single central switch.	• Single Point of Failure: If the central switch fails, the entire network goes down. • Switch Bottleneck: Performance relies heavily on the central switch's processing capacity and bandwidth limits.

	• Cost-Effective Cable Usage: Uses minimal cabling compared to a full mesh, reducing overall installation costs.	
Mesh Topology	• High Fault Tolerance: Redundant paths ensure that if one cable or node breaks, data automatically reroutes without network downtime. • No Data Collisions: Dedicated point-to-point links allow multiple devices to transmit simultaneously without congestion.	• Extremely Expensive: Requires massive amounts of cabling and hardware ports, scaling rapidly as devices are added ($\frac{N(N-1)}{2}$ links). • Complex Installation & Maintenance: Designing, routing, and configuring redundant physical links across large networks is highly complex.